DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 70

Question description

You are given a list consisting of n integers. Your task is to remove elements from the list at given positions, and report the removed elements.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$
- $1 \le p_i \le n - i + 1$

Input

The first input line has an integer n: the initial size of the list. During the process, the elements are numbered 1,2,…,k where k is the current size of the list.
The second line has n integers x1,x2,…,xn: the contents of the list.
The last line has n integers p1,p2,…,pn: the positions of the elements to be removed.

Output

Print the elements in the order they are removed.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
2 6 1 4 2
3 1 3 1 1
```

EXPECTED OUTPUT

```
1 2 2 6 4
```

Test Case 2

INPUT (STDIN)

```
6
7 2 6 1 4 2
11 3 1 3 1 1
```

EXPECTED OUTPUT

```
2 6 7 4 2 1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void build(int k,int
l,int r)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
330
```

Test Case 3

NLOC

```
37
```

Code

solution.c

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    int curSize = n;
    for (int i = 0; i < n; i++) {
        int p;
        scanf("%d", &p);
        p--;
        printf("%d%s", a[p], (i == n - 1) ? "" : " ");
        for (int j = p; j < curSize - 1; j++) {
            a[j] = a[j + 1];
        }
        curSize--;
    }
    printf("\n");
    return 0;
}
```

Output

